

📁 AI Roadmap Generator

An intelligent Next.js web application that generates personalized learning roadmaps for any topic using AI. Users can input any subject and receive a comprehensive, structured learning path tailored to their goals.

📁 Features

📁 **AI-Powered Roadmap Generation** - Generate detailed learning paths for any topic 📁 **Intelligent Structuring** - AI organizes content into logical learning phases 📁 **Comprehensive Coverage** - From beginner to advanced concepts ⚡ **Instant Generation** - Fast AI-powered responses 📁 **Fully Responsive Design** - Works seamlessly on all devices 📁 **Clean Modern UI** - Beautiful, intuitive interface 📁 **Topic Flexibility** - Support for any learning domain (tech, business, arts, etc.)

📁 Project Structure

```
├── app/                # Next.js App Router pages and components
├── app/api             # API Endpoint Connection for LLM API
├── components/         # Reusable UI components
├── public/            # Static assets
├── package.json        # Dependencies and scripts
└── README.md          # Project documentation
```

📁 Tech Stack

- **Next.js 15** - React framework for SSR and routing
- **TailwindCSS** - Utility-first styling framework
- **TypeScript** - Type safety and better development experience
- **Gemini 1.5 Flash LLM API** - Intelligent roadmap generation powered by Google's AI
- **Vercel** - Deployment and hosting platform

📁 Installation & Setup

1. Clone the repository

```
git clone https://github.com/pd241008/RoadMap-Genrator.git
cd RoadMap-Genrator
```

2. Install dependencies

```
npm install
```

3. Environment Setup

Create a `.env.local` file and add your AI service API keys:

```
# Add your AI service API key here
OPENAI_API_KEY=your_api_key_here
# or other AI service credentials
```

4. Run the development server

```
npm run dev
```

5. Open in browser

Navigate to (`http://localhost:3000`)

▣ Usage

1. **Enter a Topic:** Type any subject you want to learn (e.g., "Data Science", "Web Development", "Digital Marketing")
2. **Generate Roadmap:** Click the "Generate" button to create your personalized learning path
3. **Follow the Path:** Receive a structured roadmap with phases, milestones, and resources
4. **Customize:** Modify or regenerate roadmaps based on your specific needs

▣ Example Topics

- **Technology:** Python Programming, Machine Learning, Cloud Computing
- **Business:** Digital Marketing, Project Management, Entrepreneurship
- **Creative:** UI/UX Design, Photography, Content Writing
- **Academic:** Mathematics, Physics, Literature
- **Skills:** Public Speaking, Leadership, Time Management

▣ API Usage

The application may expose API endpoints for programmatic roadmap generation:

Endpoint:

```
POST /api/generate-roadmap
```

Request Body:

```
{
  "topic": "Machine Learning",
  "level": "beginner",
  "timeframe": "6 months"
}
```

Response:

```
{
  "roadmap": {
    "title": "Machine Learning Learning Path",
    "phases": [...],
    "estimated_duration": "6 months",
    "resources": [...]
  }
}
```

▣ Contributing

Contributions are welcome! Please feel free to submit a Pull Request.

1. Fork the project
2. Create your feature branch (`git checkout -b feature/AmazingFeature`)
3. Commit your changes (`git commit -m 'Add some AmazingFeature'`)
4. Push to the branch (`git push origin feature/AmazingFeature`)
5. Open a Pull Request

▣ License

This project is licensed under the MIT License - see the (`LICENSE`) file for details.

▣ Links

- **Live Demo:** (<https://road-map-generator.vercel.app/>)
- **GitHub Repository:**(<https://github.com/pd241008/RoadMap-Generator>)